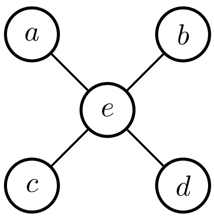
**NANYANG TECHNOLOGICAL UNIVERSITY**

**SPMS/DIVISION OF MATHEMATICAL SCIENCES**

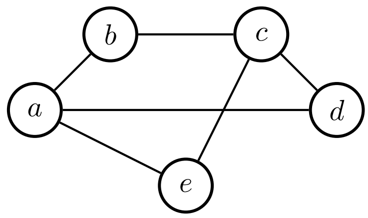
2023/24 Semester 2 MH1301 Discrete Mathematics Tutorial 9

For the tutorial in Week 10 on 28 March, let us discuss the following questions.

**Question 1. (Ex. 10.2.17, 18.)** Determine whether the graph is bipartite.




Ex. 10.2.17




Ex. 10.2.18.

**Solution:** Ex. 10.2.17. Yes, it is bipartite. Let $V_1 = \{a, b, c, d\}$ and $V_2 = \{e\}$.

Ex. 10.2.18. Yes, it is bipartite. Let $V_1 = \{a, c\}$ and $V_2 = \{b, d, e\}$.

**Question 2. (Ex. 10.2.22).** For which values of $n$ are these graphs bipartite?

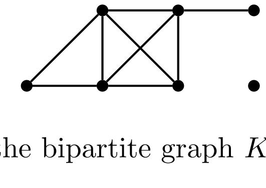
(a) $K_n$,
(b) $C_n$.

**Solution:**

Recall from lecture that a graph is bipartite if and only if its chromatic number is $\leq 2$.

(a) The chromatic number of $K_n$ is $n$. Thus only $K_1$ and $K_2$ are bipartite.
(b) The chromatic number of an odd cycle is 3 and the chromatic number of an even cycle is 2. Therefore, $C_n$ is bipartite for all $n$ even.

**Question 3. (Ex. 10.2.27).** The degree sequence of a graph is the sequence of the degrees of the vertices of the graph in nonincreasing order. For example, the degree sequence of the graph below is 4,4,4,3,2,1,0.



What is the degree sequence of the bipartite graph $K_{2,3}$? How about $K_{m,n}$?

**Solution:**

(a) Every vertex in $V_L$ is has an edge between every vertex in $V_R$. Thus the degree of every vertex in $V_L$ is 3 and the degree of every vertex in $V_R$ is 2. Thus the degree sequence is 2,2,2,3,3.

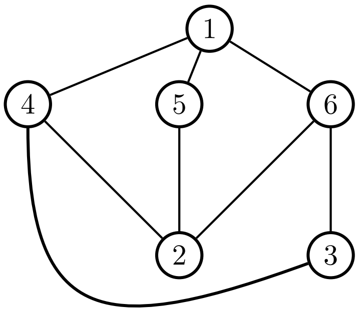
(b) Similarly the degree sequence of $K_{m,n}$ is (assuming $m \leq n$),

$$\underbrace{m, m, \ldots, m}_{n \text{ many}}, \underbrace{n, n, \ldots, n}_{m \text{ many}}.$$

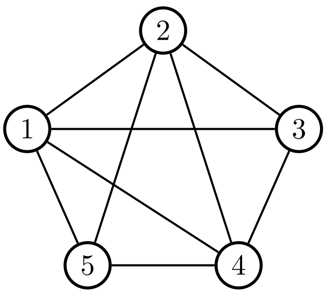
**Question 4. (Ex. 10.7.15).** Prove that removing one edge from $K_{3,3}$ leads to a planar graph. How about $K_5$ ?

**Solution:**

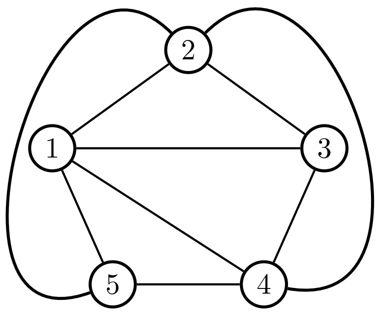
Besides drawing the graph without edges crossing, one can also argue that this graph does not have a subgraph homeomorphic to $K_{3,3}$ or to $K_5$, and invoke Kuratowski's Theorem. But here is a drawing (edge $(3, 5)$ is removed from this $K_{3,3}$):




Here is a drawing for the case of $K_5$ (edge $(3, 5)$ is removed from this $K_5$):



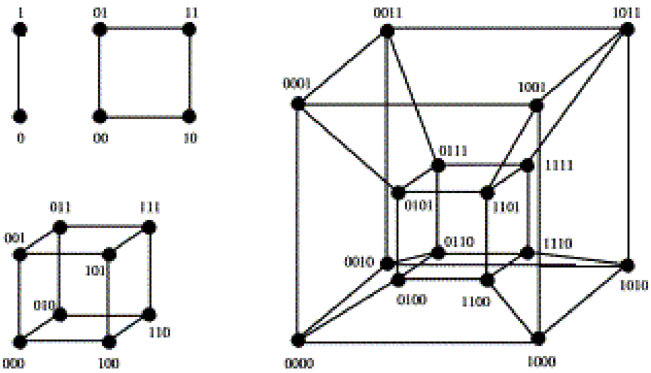

equivalently




**Question 5.** The $n$-dimensional *hypercube* $Q_n$ is a graph whose vertices are the $2^n$ strings of length $n$ over the alphabet $\{0, 1\}$, and for which two vertices are connected if and only if the associated strings differ in exactly one position. Note that $Q_n$ has $2^n$ vertices and $n/2 \cdot 2^n$ edges (the degree of all vertices is $n$).

The hypercubes $Q_1, Q_2, Q_3, Q_4$[1] look like this:

[1] $Q_2$ is a square, $Q_3$ a cube, and $Q_4$ is called a *tesseract*.




Determine the chromatic number of $Q_n$ for all $n$.

**Solution:** All $Q_n$ are bipartite (and contain edges), hence their chromatic number is 2. To show that $Q_n$ is bipartite, we partition the set of vertices $V_n$ into $V_n^0$ and $V_n^1$, where $V_n^0$ contains all vertices whose string contains an even number of 1's, and $V_n^1$ all vertices whose string contains an odd number of 1's. Clearly there is no edge between two vertices in $V_n^0$: their strings both contain an even number of 1's and with an edge they must differ in one position, which is not possible.

**Question 6. (Ex. 10.2.60. and Ex. 10.2.65).**

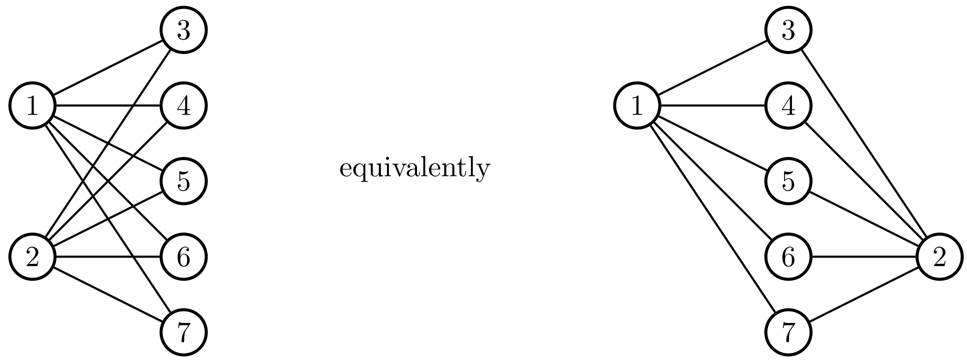
(a) Find a planar graph $G$ such that $\chi(G) = 4$.
(b) Prove that if, for a graph $G$, we remove one vertex and the edges incident to it, then the resulting graph $G'$ has chromatic number either $\chi(G)$ or $\chi(G) - 1$.

**Solution:**

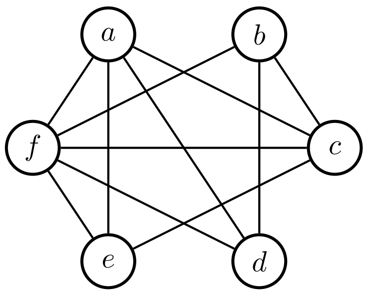
(a) $K_4$ has chromatic number 4 and is planar. Hence the four-color theorem cannot be improved to a "three-color theorem".
(b) Clearly, the chromatic number cannot increase if we remove a vertex. So the question is how much it may decrease. It is easy to see that sometimes the chromatic number does not decrease at all, for instance when we remove an endpoint vertex from the path $P_4$. The most it can decrease is, however, 1. If $G'$ had $\chi(G') = k < \chi(G) - 1$, then we would get a coloring of $G$ with $k + 1 < \chi(G)$ colors: simply color the removed vertex $v$ with a new color not used in the coloring of $G'$.

**Question 7. Ex. 10.7.1.** Can five houses be connected to two utilities without connections crossing?

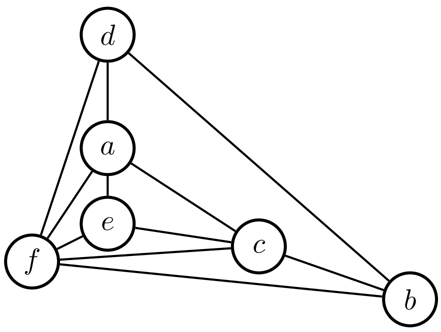
**Solution:** Yes. It can be represented by $K_{2,5}$ and $K_{2,5}$ is planar. We illustrate that with the following planar representation of $K_{2,5}$.




**Question 8. Ex. 10.7.5.** Determine whether the given graph is planar. If so, draw it so that no edge cross.




**Solution:** Here is a planar representation.




**Question 9.**

(a) Is there a planar simple connected graph with more than 3 vertices, such that removing any vertex and incident edges from it results in $\chi(G) - 1$ ? If so, find one.

(b) Is there a planar simple connected graph with 5 or less vertices, such that removing any vertex and incident edges from it results in $\chi(G)$ ? If so, find one.

**Solution:**

(a) $K_4$ is planar, and its chromatic number is 4 since every vertex adjacent to all others, so all 4 vertices need to be colored differently. Removing any vertex and its incident edges will result in $K_3$, which is 3 colorable.

(b) $C_4$ is 2-colorable. Removing any vertex from it will result in a path of length 2, which is also 2-colorable.

**Question 10.**

(a) How many edges are on a Hamilton path?

(b) Is there a connected graph with an Euler path that has the degree sequence 4,4,2,1,1?

**Solution:**

(a) A Hamilton path visits every vertex exactly once (without coming back to the first vertex), and hence has $n-1$ edges.

(b) No. Such a graph would have 5 vertices, with 2 of them connected to all other vertices. Hence there could be no vertices of degree 1. It does not matter that the graph is furthermore supposed to have an Euler path.

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## Answer.

1) Yes; Yes.
2 1 and 2; all even $n$.
3 2,2,2,3,3; $m, \ldots, m, n, \ldots, n$. ($m$ many $m$'s followed by $n$ many $n$'s.)
5) 2.
7 Yes.
8 Yes.
10) $n-1$; No.